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Active control as evidence in favor of sense of ownership in the moving Virtual Hand Illusion

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ABSTRACT

The sense of ownership, the feeling that our body belongs to ourselves, relies on multiple sources of sensory information. Among these sources, the contribution of visuomotor information is still debated. We tested the effect of active control in the sense of ownership in the moving Virtual Hand Illusion. Participants reported sense of ownership and sense of agency over a virtual arm in which we manipulated the morphological congruence of the hand and the visuomotor information. We found that congruent active control enhanced and maintained the reported sense of ownership over a hand that appeared detached from the body, but not in a morphological congruent limb. Also, incongruent active control, achieved by adding noise to the trajectory of the movement, decreased both reported sense of agency and ownership. Overall, our results are consistent with a framework in which active control acts as evidence for eliciting a sense of ownership.

1. Introduction

> "What is more important for us, at an elemental level, that the control, the owning and operation, of our own physical selves? And yet it is so automatic, so familiar, we never give it a thought." (Sacks, 1985)

The integration of multiple sources of sensory information creates and constantly updates the representation of the body in the brain (Gallagher, 2005; Metzinger, 2003). Multisensory integration, of both exteroceptive and interoceptive information, is crucial for the sense of ownership (Gallagher, 2005; Longo, Schüür, Kammers, Tsakiris, & Haggard, 2008; Makin, Holmes, & Ehrsson, 2008), the feeling that a body or body part belongs to the self (Gallagher, 2000; Longo et al., 2008).

Manipulating the sensory signals received by the body can elicit a sense of ownership over a fake body part. Such is the case of the Rubber Hand Illusion (RHI) paradigm (Botvinick & Cohen, 1998). Participants are presented with a rubber hand in an anatomically plausible position, while their real hand is hidden from their sight. Visuotactile stimulation on both the real and the rubber hand generates a sensory conflict between the felt tactile stimulation on the hidden real hand and the seen stimulation on the rubber hand. Differential integration of the visual, tactile, and proprioceptive sensory information resolves this conflict. When the stimulation is synchronous, visual information dominates, resulting in an illusion of ownership over the fake limb (Botvinick & Cohen, 1998; Makin et al., 2008; Tsakiris, 2010). However, this illusion is abolished by both temporal (Botvinick & Cohen, 1998; Perez-Marcos, Sanchez-

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Vives, & Slater, 2012) and spatial asynchrony (Kammers, Longo, Tsakiris, Dijkerman, & Haggard, 2009).

A Bayesian causal inference model has been proposed to explain this phenomenon (Kilteni, Maselli, Kording, & Slater, 2015; Samad, Chung, & Shams, 2015). In this model, the multisensory conflict is reconciled in one of two ways: either the sensory information is inferred from one common source (i.e. rubber hand), resulting in the illusion of ownership; or from two different sources (i.e. rubber and real hand), which abolishes the illusion. The likelihood of the origin of this sensory data depends on the semantic information of the hand (i.e. morphological appearance), the anatomical constraints, the proprioceptive information, the visual and tactile information of the stimulation applied, and prior expectations (Kilteni et al., 2015). In this model, the RHI occurs with congruent visual and tactile information increasing the likelihood of integrating all the information as originating from the rubber hand. This model considers the integration of tactile, visual and proprioceptive information, but does not account for the role of visuomotor information as a trigger for the sense of ownership.

In the present study, we were primarily concerned with how congruent motor information from a voluntary action and visual information of the moving limb elicit and maintain a sense of ownership. Currently, there is no consensus on the effect of active control on the sense of ownership over a fake limb. Using a moving version of the RHI (mRHI), some studies have reported that active control enhanced the sense of ownership (Kalckert & Ehrsson, 2012; Ma & Hommel, 2013; Sanchez-Vives, Spanlang, Frisoli, Bergamasco, & Slater, 2010; Slater, Spanlang, Sanchez-Vives, & Blanke, 2010; Tsakiris, Prabhu, & Haggard, 2006), while others have failed to find this effect (Dummer, Picot-Annand, Neal, & Moore, 2009; Kalckert & Ehrsson, 2014; Longo & Haggard, 2009; Riemer, Kleinböhl, Hölzl, & Trojan, 2013; Walsh, Moseley, Taylor, & Gandevia, 2011). Controlling the movement of the index finger of a rubber hand elicited a sense of ownership when the movement was synchronous, but not if it was asynchronous to the movement of the participant (Kalckert & Ehrsson, 2012; Tsakiris et al., 2006). Active synchronous control also resulted in a stronger reported sense of ownership compared to passive observation of movement and asynchronous active control (Dummer et al., 2009). This effect has also been reported in a virtual reality environment (Sanchez-Vives et al., 2010; Slater et al., 2010), and is not restricted to upper body limbs (Kokkinara & Slater, 2014). On the other hand, some studies have failed to find that sense of ownership is enhanced by the active movement of the fake limb. In a setup where participants controlled the index finger of a fake hand, active control did not increase in ownership scores with respect to externally generated passive movement (Kalckert & Ehrsson, 2014; Longo & Haggard, 2009; Walsh et al., 2011), or compared to the reported feeling of ownership after visuotactile stimulation (Dummer et al., 2009; Kalckert & Ehrsson, 2014; Longo & Haggard, 2009). Furthermore, active control failed to elicit a sense of ownership when controlling hands that have an unusual appearance such as hands with supernumerary fingers (Hoyet, Argelaguet, Nicole, & Lécuyer, 2016) or on non-hand objects (Yuan & Steed, 2010).

This lack of agreement on the effect of active control might be due to differences in the experimental setups. For instance, some studies used a rubber hand (Kalckert & Ehrsson, 2012), while others used virtual reality environments (Sanchez-Vives et al., 2010). Also, the tasks were different, ranging from single finger movements (Kalckert & Ehrsson, 2014) to full arm movements in a cognitive task (Padrao, Gonzalez-Franco, Sanchez-Vives, Slater, & Rodriguez-Fornells, 2016). Here, we propose that differences in the visual appearance of the hand could, in part, explain the differences in the effect of voluntary movement. It has been shown that the appearance of the hand needs to be congruent to the pre-existing model of the hand (Tsakiris, 2010). For instance, the illusion is abolished by non-hand shaped objects (Haans, Ijsselsteijn, & de Kort, 2008; Tsakiris & Haggard, 2005), with increasing degrees of transparency of the hand (Martini, Kilteni, Maselli, & Sanchez-Vives, 2015), when the limb has been stretched beyond a certain extent (Kilteni, Normand, Sanchez-Vives, & Slater, 2012), or with a break in body continuity (Perez-Marcos et al., 2012; Tieri, Gioia, Scandola, Pavone, & Aglioti, 2017; Tieri, Tidoni, Pavone, & Aglioti, 2015a). Specifically, seeing the hand detached from the virtual body decreased sense of ownership over static limb (Tieri et al., 2017, 2015a; Tieri, Tidoni, Pavone, & Aglioti, 2015b) and passive observation of movement (Tieri et al., 2015b). However, whether active control of the movement helps maintain the sense of ownership of a hand that appears detached from the body remains to be tested. We reasoned that, if active control was relevant for the sense of ownership, the decrease in sense of ownership due to morphological incongruence should not occur.

The sense of agency refers to the subjective experience of being the source of the action (Haggard, 2017) and its consequences in the environment (Caspar, Cleeremans, & Haggard, 2015; Moore & Fletcher, 2012) and is closely related to action awareness and action planning (de Vignemont, 2011). In a voluntary action, motor commands sent to the muscles generate an efference copy that will be used to predict the future bodily states (Wolpert, 1997). Sensory information predicted from the motor command is compared to the actual sensory feedback from the body and the environment (Farrer & Frith, 2002); when these two match, sense of agency arises (Blakemore, Wolpert, & Frith, 1998). Conversely, a visual or temporal mismatch abolishes sense of agency (Blakemore, Wolpert, & Frith, 2002). This comparator model highlights the importance of visuomotor effects of the movement for the sense of agency (see Haggard, 2017 for review). However, a sense of agency can also be present without any actual movement (Tieri et al., 2015b; Wegner, Sparrow, & Winerman, 2004). In the absence of motor commands, passively observing a limb moving generates vicarious agency over the movements of the limb (Pezzetta, Nicolardi, Tidoni, & Aglioti, 2018; Tieri et al., 2015b; Wegner et al., 2004).

In the present study, we assessed the role of active control as evidence in favor of a sense of ownership over the virtual hand. We used a variation of the mVHI paradigm in which participants controlled the virtual hand during a goal-directed task. We hypothesized that visual appearance plays a role in the effect of active control. If that is case, active control would enhance the sense of ownership over the virtual limb when the evidence was lower. We used a manipulation in which the hand appears detached from the body (Perez-Marcos et al., 2012; Tieri et al., 2017; 2015a, 2015b). Additionally, we tested the importance of movement congruence to the movement of the participants in sense of ownership. Overall, we aim to understand the role of congruent active control in sense of ownership.

2. Materials and methods

2.1. Participants

A total of 44 volunteers (29 females; age range = 18–45; mean age = 28.2 ± SD = 7.5 years) took part in the study. Participants signed up through a public online form. Those that declared having no neurological conditions and to be right-handed were contacted to participate. All subjects were naïve to the purposes of the study. Seven participants were excluded from the analysis: four participants were excluded due to technical issues (e.g. the tracking of the virtual hand failed during a dynamic trial) and another three were excluded as they failed to follow the instructions (e.g. moved their hand before or during the static trials). In all the cases, participants performed the experiment until completion, but the data were excluded from the analyses. Thus, all analyses were performed on a sample of 37 participants (23 females, age range = 18–45; mean age = 27.6 ± SD = 6.7 years). Participants were tested for handedness using the Edinburgh Handedness Inventory (Olfield, 1971) (mean = 75.86 ± SD = 24.83). 73% never had any previous contact with virtual reality and 55% declared not to be regular video game players.

The experimental protocol was approved by the Institutional Review Board at the Champalimaud Foundation and was carried out in accordance with the ethical standards of the Declaration of Helsinki. All participants provided written informed consent prior to participation. At the end of the experiment, participants were paid €15 in a voucher and were debriefed. The whole experiment lasted approximately 30 min.

2.2. Virtual environment and apparatus

Participants were immersed in the virtual environment using an Oculus Rift DK2 (Oculus VR, LLC) head-mounted display and saw the virtual environment from a first-person perspective. The virtual environment consisted of a large room with the same size and appearance as the experimental room. All components of the virtual environment were designed using 3DS Max 2015 (Autodesk, Inc) and implemented in Unity 3D Engine (Unity Technologies SF). The room contained a table, which was a 1:1 replica of the real one, with the same shape, size (100 cm × 36 cm × 77 cm), and position. Both the real and the virtual table had three circles on top that acted as lights in the virtual environment. The virtual and the real scenes were carefully aligned by measuring the location of the positional tracking camera of the Oculus Rift (near-infrared CMOS sensor) in the real room with the same point in the virtual scene.

Participants controlled a gender-matched right arm in an anatomical congruent position that was collocated with their real, hidden right arm (Fig. 1A, B). The virtual arm was controlled using a LEAP motion controller (Leap Motion, Inc). This device captures the kinematics of the real hand of the participants and transforms it online into the virtual hand movement. The virtual and real hands were carefully aligned by measuring the LEAP Motion controller in the experimental room and using the same location in the virtual world as the origin for the hand models. We used two different hand models: one that appeared attached to the body and one detached without a forearm (Fig. 1B). These hand models were provided by the LEAP motion SDK. Throughout the whole experiment, the trajectories of the participants' hand were logged using the LEAP motion. Due to technical limitations, we did not show a virtual body in the environment. However, participants were always requested to look at the hand, which minimized the effect of this constraint.

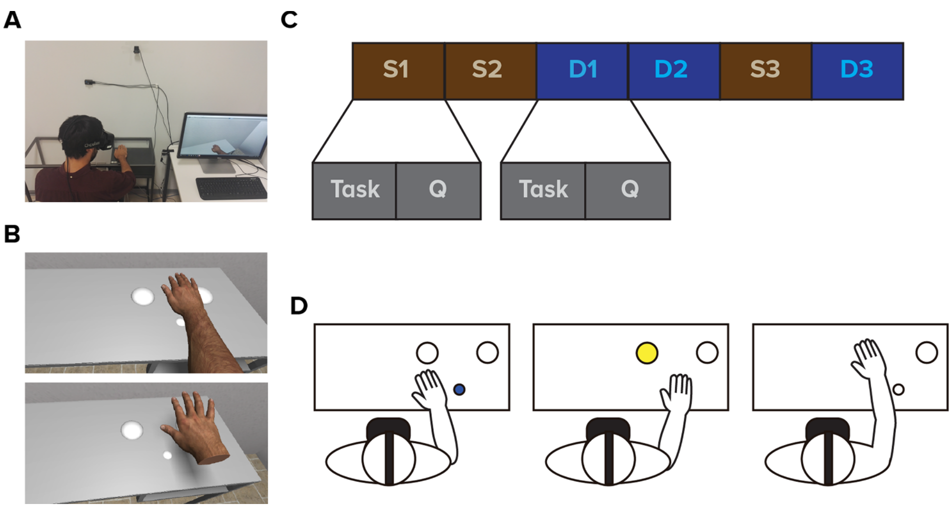

Fig. 1. Virtual reality environment and experimental design. (A) Virtual and real set up. (B) Male virtual hand model for full arm and detached hand. Hand models were gender-matched. (C) Experimental design. Participants underwent four blocks consisting in a total of six conditions, three static (S1, S2, S3) and three dynamic (D1, D2, D3). Each condition consisted of a task followed by the subjective report of the illusion. (D) In the dynamic conditions, participants performed a reaching task consisting of 25 reaching movements.

2.3. Procedures

Upon arrival, participants were asked to read and sign a consent form. The experimenter verbally explained the instructions of the experiment to them. Participants were seated in front of the table and requested to place their right hand on top of it. We instructed the participants to refrain from moving the hand from this point until told otherwise. The experimenter fitted the head mounted display and the experiment started.

Each participant experienced a total of six conditions. Each condition consisted of a task, followed by the subjective report of the sense of ownership and sense of agency (Fig. 1C). The task differed depending on the type of condition. Participants either observed a static right hand without attempting to move it (static conditions) or controlled the hand in a goal-directed task that required performing reaching movements (dynamic conditions). When the task was completed the right hand disappeared, and participants verbally reported their subjective sense of ownership and sense of agency by responding to the questionnaires (see Section 2.6). After this, the next condition began. Participants did not remove the headset at any point during the experiment. At the end of all the conditions, participants were asked for their overall qualitative experience of the experiment, were paid, and dismissed.

2.4. Experimental design

We divided the conditions into four blocks (Fig. 1C). The first block consisted of two static conditions (S1 and S2): one in which the arm appeared attached to the body (*full arm static*) and one in which the forearm was missing (*detached hand static*). These two conditions were counterbalanced within this block across participants. The second block consisted of two dynamic conditions (D1 and D2): one in which the arm appeared to be attached (*full arm dynamic*) and one in which the forearm was missing (*detached hand dynamic*). These two conditions were counterbalanced within this block across participants. To avoid artifacts coming from participants having controlled the virtual arm, S1 and S2 were always used before D1 and D2. The third block consisted of a single static condition, which we labeled *full arm static - post* (S3). This condition was used as a control for the effects of active control on a subsequent static condition, after participants were allowed to control the virtual hand. Finally, the last block consisted of a single dynamic condition (D3) in which participants controlled a full arm where the movement was incongruent to the actual movement of the participant (*incongruent movement* condition).

2.5. Details of the conditions

2.5.1. Static conditions

During the static tasks, participants were asked to keep their hand still and stare at it during 60 s. To ensure that no hand movement could be seen or performed during these trials, the position of the hand model was not updated and the participant was carefully monitored by the experimenter during the whole task.

2.5.2. Dynamic conditions

The dynamic tasks consisted of performing a set of reaching movements towards a target light. A schematic of the task is depicted in Fig. 1D. Before the task started, participants were requested to lift their hand from the table and abstain from placing it back until told otherwise, as to avoid tactile information from the real arm. First, the small central light turned on blue. Participants were required to place their hand on top without touching it. This caused one of the two large target lights to turn on yellow, determined randomly. The subject had to reach towards the newly illuminated light and to turn it off within the specified time of five seconds. If performed correctly, the target light would change to green; otherwise, it changed to red. After 0.5 s, the blue central light turned on again and the process was repeated for a total of 25 times. Each dynamic task lasted approximately three minutes.

2.5.3. Detached arm condition

Previous studies reported that a break in body continuity negatively affects sense of ownership and vicarious agency over a virtual arm at rest or during passive observation of the moving arm (Perez-Marcos et al., 2012; Tieri et al., 2017; 2015a). To manipulate sense of ownership over the virtual hand, we used a virtual hand model that appeared detached from the body with a clean cut at the wrist (Fig. 1B).

2.5.4. Incongruent movement condition

In the *incongruent movement* condition the visual information from the movement of the virtual hand was incongruent to the action performed by the participant. To do this, we added noise to the trajectory of the rubber hand.

The virtual arm position, $\hat{p}_i$, is defined by a combination of the current hand location with the virtual hand location in the previous frame, and some noise defined by the equation:

$$\hat{p}_i = \lambda p_i + (1 - \lambda)(\hat{p}_{i-1} + (p_i - p_{i-1}) + r)$$

where $\hat{p}_{i-1}$ is the previous virtual hand location; p_i and p_{i-1} are the present and previous real hand locations; and r is Gaussian random noise of mean 0 and standard deviation σ; and the parameter λ determines the weight of the present location in favor of the noisy virtual location. In our experiment $\sigma = 0.015$ m and $\lambda = 0.9$. This means that 90% of the virtual hand movement was based on the current sample (i.e. with added latency), and the other 10% was composed of the actual delta (previous vs present virtual hand location) and random noise to the previous virtual hand location.

Table 1

Statements of the questionnaires for both static and dynamic conditions. We presented two questionnaires depending on the type of condition (static or dynamic). Statements regarding sense of ownership were the same in both static and dynamic conditions, while sense of agency differed.

	Category	Condition Type	Statement
1	**Sense of ownership**	Static and Dynamic	I felt as if I were looking at my hand, rather than a virtual hand
2	**Sense of ownership**	Static and Dynamic	I felt as if the virtual hand was my hand
3	**Sense of ownership**	Static and Dynamic	It seemed like the virtual hand belonged to me
4	**Sense of ownership**	Static and Dynamic	It seemed that the virtual hand was part of my body
5	**Sense of ownership - control**	Static and Dynamic	I felt as if I had more than one right hand
6	**Sense of ownership -control**	Static and Dynamic	I felt like my real hand was turning virtual
7	**Similar**	Static and Dynamic	I felt as if the virtual hand physically resembled my real hand in terms of shape, freckles and other features
8	**Sense of agency**	Static	I felt that I could control the virtual hand if I wanted to
9	**Sense of agency**	Static	I felt that I started to move my hand, the virtual hand would obey my will
10	**Sense of agency – control**	Static	I had the feeling of forgetting my own hand, focusing only on the virtual hand
11	**Sense of agency**	Dynamic	I felt that I could control the virtual hand
12	**Sense of agency**	Dynamic	I felt that the movements of the virtual hand were cause by me
13	**Sense of agency**	Dynamic	I felt as if the virtual hand was obeying my will
14	**Sense of agency**	Dynamic	I felt that I controlled the virtual hand as if it was part of my body
15	**Sense of agency – control**	Dynamic	I felt as if the virtual hand was controlling my hand
16	**Sense of agency – control**	Dynamic	I had the feeling of forgetting my own hand, focusing only on the movement of the virtual hand
17	**Sense of agency – control**	Dynamic	I felt as if the virtual hand caused the movement of my hand
18	**Outcome agency**	Dynamic	I felt as if the lights were obeying my will
19	**Outcome agency**	Dynamic	I felt that the movement of my hand was turning off the lights on the table
20	**Outcome agency – control**	Dynamic	I felt as if the lights changed at random
21	**Sense of location**	Static and Dynamic	It seemed like my hand was in the location of the virtual hand
22	**Sense of location**	Dynamic	It seemed as if the movement of my hand was located where the virtual hand was moving

2.6. Questionnaires

To assess the participant's subjective experience, we used two questionnaires adapted from previous studies (Botvinick & Cohen, 1998; Kalckert & Ehrsson, 2012), one for each type of condition (i.e. static or dynamic). The participants rated their subjective experience on a seven-point Likert scale (1 = totally disagree, 4 = neutral, 7 = totally agree).

For the static conditions, we used an eleven-item questionnaire that assessed ownership (statements 1–4), sense of agency (statements 8–9), and sense of location (statement 21). We also used control statements of ownership (statements 5–6) and agency for static conditions (statement 10). For the dynamic conditions, the questionnaire consisted of nineteen items that assessed sense of ownership (statements 1–4), sense of agency in dynamic condition (statements 11–14) and sense of location (statement 21–22). Additionally, we used control statements of ownership (statements 5–6) and agency for the dynamic conditions (statement 15–17). The dynamic conditions also contained items regarding the perceived agency over the consequences of the action (hereafter, outcome agency) (statements 18–19), and the respective control statement (statement 20). Statements assessing sense of agency were different between static and dynamic conditions, to avoid priming effects from the statements referring to an active control in the first static block. The control statements served as controls for task compliance and suggestibility effects. We also assessed the subjective similarity between the real and the virtual hand (statement 7) (Table 1).

The experimented manually recorded the responses on paper and using the Likert.m Matlab function, modified to present the statements in a counterbalanced order across all conditions and participants.

2.7. Data handling

All data were analyzed using Matlab 2014b.

The questionnaires included more than one statement for sense of ownership, sense of agency (static), sense of agency (dynamic), sense of agency (outcome), and their respective control statements (see Section 2.6). To avoid the artifacts related to pseudoreplication, for each participant we calculated the individual mean score for each category (e.g. sense of ownership) in each condition (e.g. *full arm static*). Thus, we obtained a total of 37 individual scores for each category and condition. Using the Shapiro-Wilk test, the individual ownership scores were not found to follow a normal distribution, while the individual agency scores did. Therefore, we used non-parametric tests for the remaining analysis. We used a Friedman ANOVA to compare across all the conditions and the Wilcoxon signed-rank test for pairwise comparisons.

We compared the sense of ownership and the sense of agency statements to their respective control statements in each condition to test for task compliance and suggestibility effects. Also, in accordance to our hypotheses, we compared the ownership scores in the following pairs of conditions: *full arm static* against *full arm dynamic* and *detached hand static* against *detached hand dynamic*, to assess the effect of active control; *full arm static* against *detached hand static* and *full arm dynamic* against *detached hand dynamic*, to assess the effect of morphological congruence; and *full arm dynamic* against *incongruent movement* to assess the importance of movement congruence. We also compared the following pairs of conditions for the sense of agency: *full arm static* against *detached hand static* and *full arm dynamic* against *detached hand dynamic*, to assess the effect of morphological congruence; and *full arm dynamic* against *incongruent movement* to assess the importance of movement congruence. Finally, to assess the effects of morphological incongruence and movement incongruence on outcome agency, we compared *full arm dynamic* against *detached hand dynamic* and *full arm dynamic* against *incongruent movement*. Thus, in total, we performed twelve pairwise comparisons, each with a Bonferroni corrected p-value of 0.004.

Additionally, we reasoned that participants might be affected differently by the manipulations depending on their basal level of ownership. For this reason, we analyzed the reported ownership separating the participants into three groups according to their individual ownership score: low (i.e. individual ownership score ≤ 3), medium (i.e. individual ownership score > 3 and < 5), and high (i.e. individual ownership score ≥ 5). We compared participants within each group in the following pairs of conditions using the Wilcoxon signed-rank test: *full arm static* against *full arm dynamic* and *detached hand static* against *detached hand dynamic*, to assess the effect of active control movement; *full arm static* against *detached hand static* and *full arm dynamic* against *detached hand dynamic* to assess the effect of morphological congruence; and *full arm dynamic* against *incongruent movement* to assess the importance effect of movement congruence. Each of these five pairs of conditions was tested for the three groups with different basal individual ownership scores, giving a total of 15 tests, each with a Bonferroni corrected p-value of 0.003.

Lastly, we assessed the correlation between the changes in sense of agency and sense of ownership in two conditions. To this end, we calculated the differences in the individual agency scores and ownership scores in two conditions and calculated the correlation in these differences using the Spearman ρ correlation coefficient.

3. Results

3.1. Sense of ownership and sense of agency over a virtual hand

We first assessed whether the statements for the sense of ownership and sense of agency (dynamic and static) were higher their respective control statements for each experimental condition (see Section 2.6).

Participants reported a sense of ownership after seeing the arm at rest connected to the body (*full arm static*) (median = 4.5 ± 1.45; see Table 2 for the median ownership scores in all conditions). In this condition, the reported sense of ownership was higher than the respective control statements (Wilcoxon signed-rank; Z = 4.56, $p < 0.001$). In the *detached hand static* condition, the median reported sense of ownership was below the neutral score (score = 4) (median = 3.25 ± 1.53) and was higher than the control statements (Wilcoxon signed-rank; Z = 3.02, $p = 0.002$). The *full arm static - post* condition (median = 4.25 ± 1.54) was also significantly different from the control statements (Wilcoxon signed-rank; Z = 5.15, $p < 0.001$) (Fig. 2A).

The agency scores in the *full arm static* (median = 6 ± 1.45; see Table 2 for a summary of the median agency scores) were higher than the respective control statements (Wilcoxon signed-rank; Z = 3.12, $p < 0.001$). In the *detached hand static* condition, agency scores (median = 5 ± 1.83) showed no significant difference compared to the control statements (Wilcoxon signed-rank; Z = 1.14, $p = 0.15$). In the *full arm static - post*, participants reported agency scores (median = 6 ± 0.84) that were significantly different from the control statements (Wilcoxon signed-rank; Z = 3.66, $p < 0.001$) (Fig. 2C).

In the dynamic conditions, participants reported feeling ownership over the virtual hand in *the full arm* (median = 4.75 ± 1.39) and in the *detached hand* (median = 4.25 ± 1.34). In both conditions, ownership scores were higher than in the control statements (Wilcoxon signed-rank; Z = 4.65, $p < 0.001$, and Z = 4.92, $p < 0.001$). In the *incongruent movement* condition, the median ownership score was below the neutral score (score = 4) (median = 3.5 ± 1.62). The scores in this condition were higher than in the respective control statements (Wilcoxon signed-rank; Z = 3.73, $p < 0.001$) (Fig. 2B).

Participants also reported a sense of agency over the virtual arm movements in the *full arm dynamic* (median = 6.25 ± 0.85) and *detached hand dynamic* (median = 6.25 ± 0.72), which were higher than in their respective control statements (Wilcoxon signed-rank; Z = 5.30, $p < 0.001$; and Z = 5.28, $p < 0.001$, respectively). In the *incongruent movement* condition, the sense of agency (median = 5.5 ± 1.12) was higher than the control statements (Wilcoxon signed-rank; Z = 5.18, $p < 0.001$) (Fig. 2D).

The subsequent analyses are aimed at understanding whether the different manipulations caused a difference in the reported ownership and agency scores in the different conditions.

Table 2

Median scores for each condition and for each category of statements. Means of all the statements of a category (e.g. sense of ownership) were calculated for each individual. Using these individual scores, we calculated the median scores of the sample for each category in each condition.

	Full arm static	Detached hand static	Full arm dynamic	Detached hand dynamic	Full arm static - post	Incongruent movement
Ownership	4.5	3.25	4.75	4.25	4.25	3.5
Ownership Control	3	2.5	2.5	2.5	2.5	2.5
Agency (static)	6	5	n/a	n/a	6	n/a
Agency (static) Control	5	4	n/a	n/a	5	n/a
Agency (dynamic)	n/a	n/a	6.25	6.25	n/a	5.5
Agency (dynamic) Control	n/a	n/a	3	3	n/a	3
Outcome Agency	n/a	n/a	6	6	n/a	6
Location	7	6	6.5	6	7	6

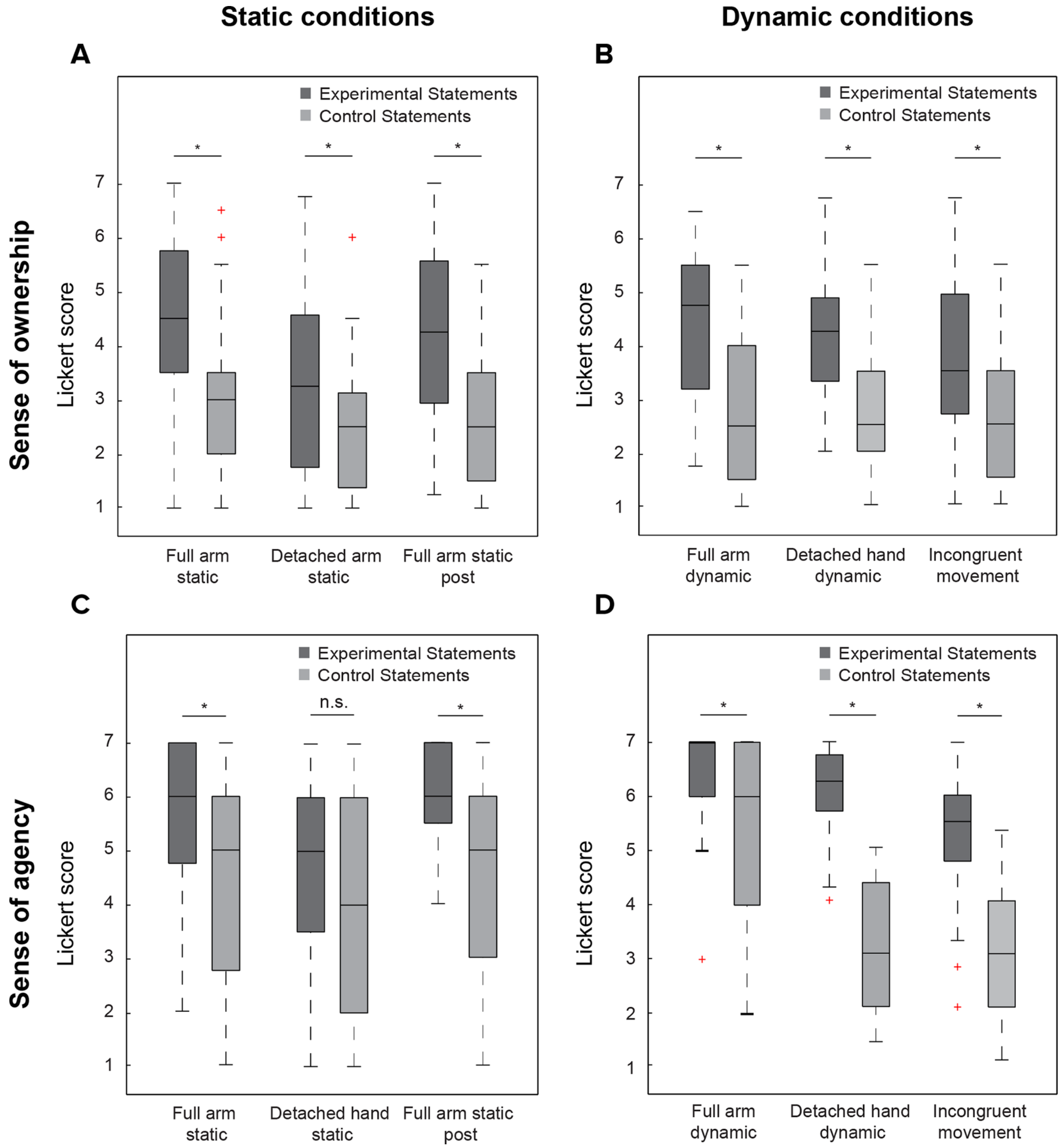

Fig. 2. Reported sense of ownership and sense of agency scores and their respective control scores. (A) Ownership scores in the static conditions (B) Ownership scores in the dynamic conditions (C) Agency scores in the static conditions (D) Agency scores in the dynamic conditions. The middle quartile indicates the median value and the whiskers indicate the most extreme values that are not considered outliers. $^{*}p < 0.05$.

3.2. Morphological incongruence decreases reported sense of ownership and agency in static, but not in dynamic conditions

We assessed whether seeing the hand detached from the body affected the reported sense of ownership and the sense of agency and if active control made a difference in this effect. Previous studies had already reported a decrease in sense of ownership and vicarious agency over a static discontinuous limb (Tieri et al., 2015b). We hypothesized that, if active control is relevant the reported sense of ownership, this decrease in reported ownership and agency would be absent when participants actively controlled the movements of the virtual hand.

Ownership scores were significantly different across all the conditions (Friedman test; $\chi^2 = 31.45$ (df = 5, n = 37), $p < 0.001$). To test for our hypothesis, we compared the ownership scores between the *full arm* conditions and their corresponding morphological incongruent conditions (i.e. *detached hand*). In the absence of movement, the ownership scores in the *detached hand* were lower than the scores in the *full arm static* condition (Wilcoxon signed-rank; Z = 4.21, $p < 0.001$) (Fig. 3A). However, we found no significant difference between the ownership scores in the *full arm dynamic* and the *detached hand dynamic* (Wilcoxon signed-rank; Z = 2.02,

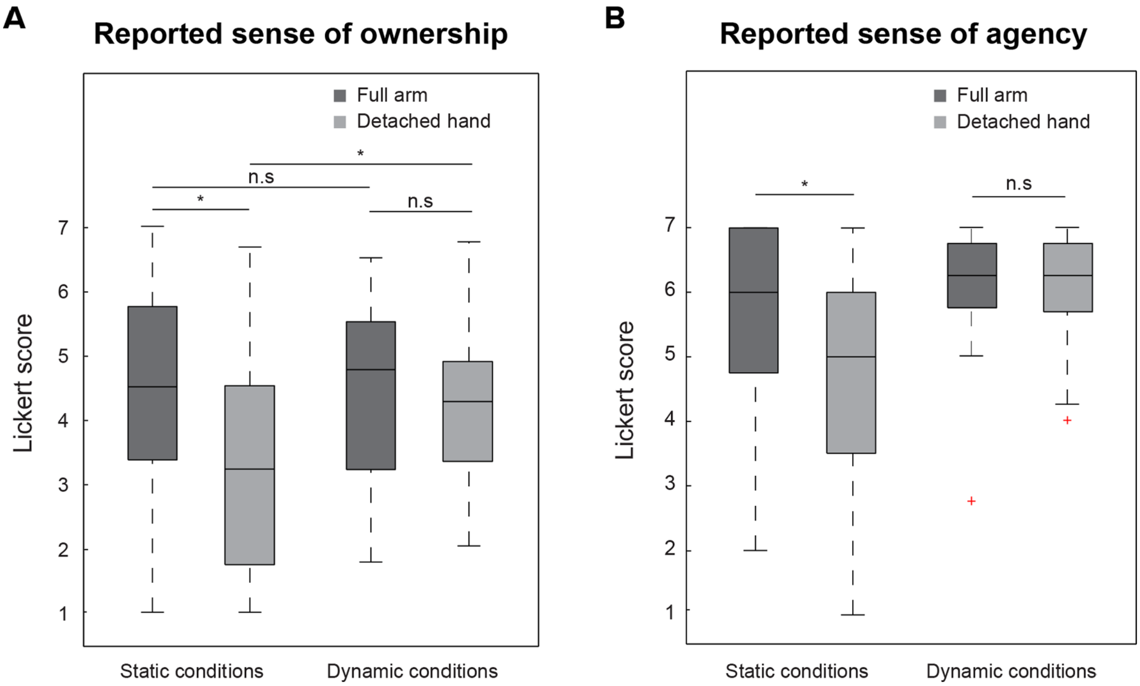

Fig. 3. Effect of movement and morphological incongruence in reported ownership and agency scores. (A) Active control did not increase reported ownership when comparing the full arm static and the full arm dynamic conditions ($p = 0.51$, Wilcoxon signed rank test). Seeing the hand detached caused a decrease in reported ownership scores in the static conditions ($p < 0.001$, Wilcoxon signed rank test), but not in the dynamic conditions ($p = 0.04$, (not significant with Bonferroni correction), Wilcoxon signed rank test). Active control increases reported ownership scores in detached hand conditions ($p < 0.001$, Wilcoxon signed rank test). (B) Seeing the hand detached decrease reported perceived agency in the static conditions ($p = 0.002$, Wilcoxon signed rank test), but it did not decrease agency scores in dynamic conditions ($p = 0.44$, Wilcoxon signed rank test). The middle quartile indicates the median value and the whiskers indicate the most extreme values that are not considered outliers. * The Bonferroni corrected p-value for these comparisons was 0.004.

$p = 0.04$; not significant with Bonferroni correction) (Fig. 3A).

Agency scores were different across the three static conditions (Friedman test; $\chi^2 = 25.29$ (df = 2, n = 37), $p < 0.001$) and the three dynamic conditions (Friedman test; $\chi^2 = 21.53$ (df = 2, n = 37), $p < 0.001$). We tested for differences in the agency scores according to the planned pairwise comparisons (see Section 2.7). In static conditions, the agency scores in the *detached hand* condition were lower than in the *full arm* condition (Wilcoxon signed-rank; $Z = 3.03$, $p = 0.002$) (Fig. 3B). However, under congruent active control (i.e. dynamic conditions), the agency scores in the *full arm dynamic* condition were not found to be significantly different to the *detached hand dynamic* agency scores (Wilcoxon signed-rank; $Z = 0.77$, $p = 0.44$) (Fig. 3B).

These findings are consistent with the idea that active control over the virtual hand movement plays a differential role in the effect of morphological incongruence in the reported sense of ownership and sense of agency.

3.3. *Congruent active control does not increase reported sense of ownership in full arm, but it does in detached hand conditions*

To assess the effect of active control on reported sense of ownership, we compared the ownership scores between the static and dynamic conditions. We hypothesized that if active control contributes to the sense of ownership, we would see an increase in the reported ownership scores for both *full arm* and *detached hand* conditions.

Sense of ownership scores were significantly different across all six conditions (Friedman test; $\chi^2 = 31.45$ (df = 5, n = 37), $p < 0.001$). To test for our hypothesis, we compared the ownership scores between the static conditions and their corresponding dynamic condition. We found no significant differences in reported sense of ownership when comparing *full arm static* and *full arm dynamic* (Wilcoxon signed-rank; $Z = 0.66$, $p = 0.51$). However, active control resulted in an increase in the ownership scores in the morphological incongruent conditions (*detached hand static* vs *detached hand dynamic*) (Wilcoxon signed-rank; $Z = -3.38$, $p < 0.001$) (Fig. 3A).

These results show that active control can increase the reported sense of ownership, but this effect occurs only in conditions when controlling a morphological incongruent virtual limb.

3.4. *Participants respond differently to each manipulation depending on their reported ownership*

So far, our results suggest that the effect of active control on reported sense of ownership is affected by the morphological

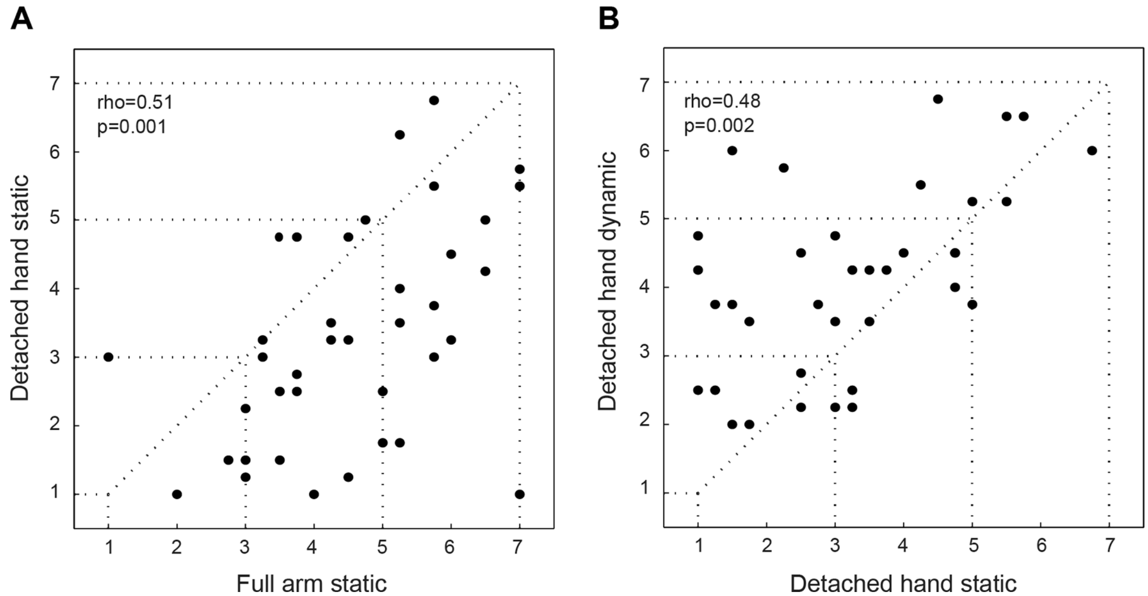

Fig. 4. Correlations of reported ownership scores. (A) Reported ownership in the full arm static and detached hand static were significantly correlated (Spearman $\rho = 0.51$, $p = 0.001$). Participants in the high ownership group ($n = 16$) were significantly different in the medians between both conditions ($p < 0.001$, Wilcoxon signed rank test), while participants in the medium ($n = 15$) and low ownership ($n = 6$) did not show any significant difference ($p > 0.4$, Wilcoxon signed rank test) (B) Reported ownership in the detached hand static and detached hand dynamic were significantly correlated (Spearman $\rho = 0.48$, $p = 0.002$). Participants in the low ownership group ($n = 18$) were significantly different between both conditions ($p < 0.001$, Wilcoxon signed rank test), while participants in the medium ($n = 13$) and high ownership ($n = 6$) groups did not show a significant difference ($p = 0.03$ (not significant under Bonferroni correction) and $p = 0.49$, respectively).

appearance of the limb. We tested whether participants responded differently to the manipulations depending on their ownership levels. For this purpose, we divided the participants into three groups according to a different baseline in each pairwise comparison (see Section 2.7), and tested for differences within each group.

In the static conditions, reported ownership was significantly different in the *full arm* and *detached hand* conditions (Fig. 3A). Individual ownership scores showed a positive correlation between these two conditions (Spearman $\rho = 0.51$, $p = 0.001$) (Fig. 4A). When comparing reported sense of ownership in the different subgroups of participants, using the *full arm static* as a baseline, seeing the hand detached from the body caused a decrease in the high ownership individuals (Wilcoxon signed-rank; $Z = 41$, $p < 0.001$, $n = 16$), but not the medium and low (Wilcoxon signed-rank; $p = 0.04$, not significant under Bonferroni, $n = 15$ and 0.53, $n = 6$, respectively) (Fig. 4A).

Active control only showed an effect in sense of ownership in morphologically incongruent conditions (*detached hand static* against *detached hand dynamic*) (Fig. 3A). These conditions were also correlated (Spearman $\rho = 0.48$, $p = 0.002$) (Fig. 4B). Furthermore, the analysis of the subgroups, using the *detached hand static* condition as the baseline, revealed that active control caused a significant increase in low (Wilcoxon signed-rank; $Z = -3.37$, $p < 0.001$, $n = 18$), but not in medium or high ownership groups (Wilcoxon signed-rank; $p = 0.03$, not significant under Bonferroni correction, $n = 13$ and $p = 0.49$, $n = 6$, respectively) (Fig. 4B). The remaining comparisons can be found in the Supplementary Materials (Fig. S1).

These results show that active control and morphological incongruence have different effects in different subgroups of participants, depending on the experimental conditions that are used as the baseline.

3.5. *Incongruent active control decreases sense of agency and sense ownership*

Up to this section, we have shown that active control is relevant in the sense of agency and the sense of ownership. We further hypothesized that, for active control to be relevant for the sense of ownership, the movement needed to be congruent to the participant's real movements. We thus used the *incongruent movement* condition as a manipulation in which noise is added to the mean trajectory of the virtual hand (see Section 2.5.4).

Sense of ownership was significantly different across all six conditions (Friedman test; $\chi^2 = 31.45$ ($df = 5$, $n = 37$), $p < 0.001$) and for sense of agency in the three dynamic conditions (Friedman test; $\chi^2 = 21.53$ ($df = 2$, $n = 37$), $p < 0.001$). Reported agency scores in the *incongruent movement* condition were significantly reduced when compared to the *full arm dynamic* condition (Wilcoxon signed-rank; $Z = 4.09$, $p < 0.001$). This was also the case for the reported ownership scores (Wilcoxon signed-rank; $Z = 3.64p < 0.001$) (Fig. 5A). These results are consistent with the idea that the movement needs to be congruent to the movement of the real hand is needed to elicit a sense of agency and ownership over the virtual hand.

We also measured the correlation of the change in individual ownership scores and the change in individual agency scores

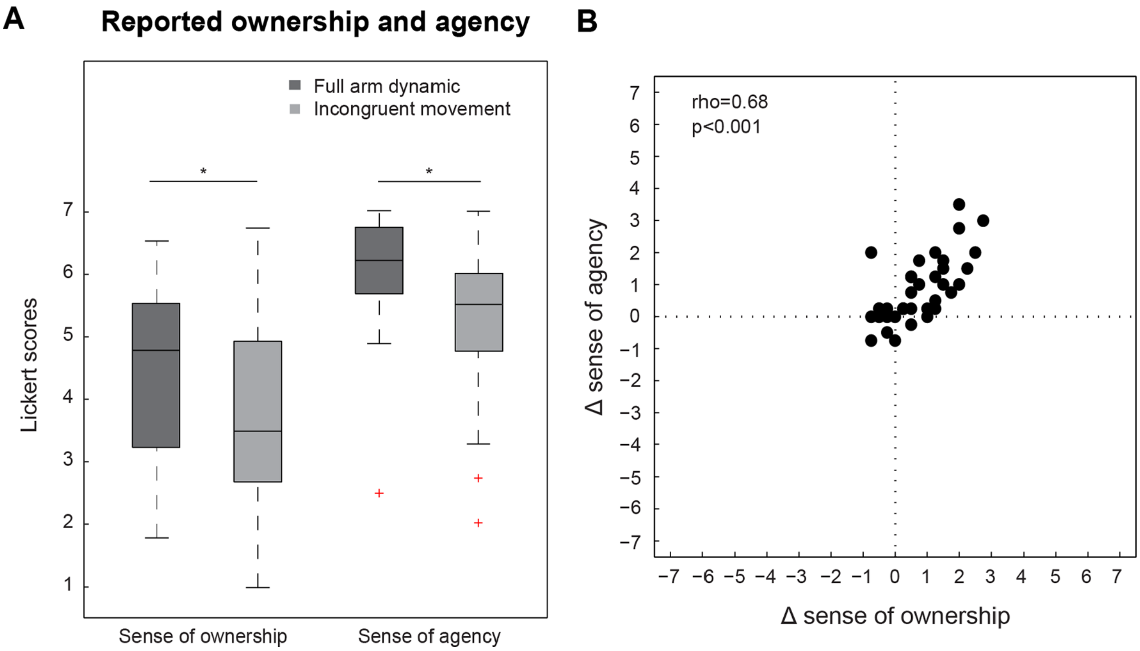

Fig. 5. Effect of incongruent movement in reported sense of ownership and sense of agency. (A) There is a significant decrease in reported sense of ownership ($p < 0.001$, Wilcoxon signed rank test) and in reported sense of agency ($p < 0.001$, Wilcoxon signed rank test). The middle quartile indicates the median value and the whiskers indicate the most extreme values that are not considered outliers. * The Bonferroni corrected p-value for these comparisons was 0.004. (B) Correlation between the difference in ownership and the difference in agency scores, when comparing full arm dynamic and incongruent movement conditions. These show a strong correlation (Spearman $\rho = 0.68$, $p < 0.001$).

between the *full arm dynamic* and *incongruent movement* conditions. The changes in both categories are strongly correlated within participants (Fig. 5B). This suggests that incongruent hand movement, in each participant, has a similar detrimental effect on both the senses of ownership and agency. This correlation is less clear when comparing the *full arm* conditions to the *detached hand* conditions in static and dynamic scenarios (Fig. S2).

4. Discussion

We here report two main findings on the effect of visuomotor information on the sense of ownership in the moving Virtual Hand Illusion (mVHI) paradigm. First, we show that morphological incongruence did not change the reported sense of ownership under active control, contrary to what is seen in static conditions. Secondly, we found that active control increased the reported sense of ownership, but only when participants controlled a detached form of the hand, in which sense of ownership was significantly lowered. Taken together, our results show that active control of the movements of the virtual hand enhanced and maintained reported ownership over the virtual hand. However, this effect was contingent on the available sensory information in favor of sense of ownership.

Seeing a static hand with the missing forearm has been reported to negatively affect the reported sense of ownership in the absence of any additional sensory information (Tieri et al., 2017, 2015a, 2015b) or after visuotactile stimulation (Perez-Marcos et al., 2012). In the absence of active control, we also found a significant decrease in the reported sense of ownership over the hand appears detached from the body (i.e. *detached hand static*) compared to *full arm static* condition. The integration of all available sensory information is critical for eliciting a sense of body ownership and is not restricted to visuotactile information (Kalckert & Ehrsson, 2012), as visuomotor information can also add evidence of feelings of ownership (Synofzik, Vosgerau, & Newen, 2008). Here, we found no significant change in the sense of ownership between the *full arm* and the *detached hand* while performing a goal-directed task. Thus, active control circumvents the decrease in reported ownership caused by seeing a morphological incongruent limb. This suggests that, when available, visuomotor information plays a critical role in maintaining a sense of ownership over a limb that appears detached from the body. To our knowledge, this is the first study to report this effect.

In this study, we also tested whether active control could enhance sense of ownership over a virtual limb. While some studies have reported that active control increased the sense of ownership (Dummer et al., 2009; Kalckert & Ehrsson, 2012), others failed to see this effect (Kalckert & Ehrsson, 2014; Walsh et al., 2011). We hypothesized that this could result, in part, from the differences in the morphological appearance of the hand, since it is known to play a crucial role in the RHI paradigm (Kilteni et al., 2015). We used the morphological incongruent form of the hand (*detached hand*) to test for this hypothesis. In our setup, the information provided to the participants in *full arm static* condition (i.e. first-person perspective, anatomical plausibility, realistic appearance of the hand) was sufficient to drive the illusion of ownership (Costantini & Haggard, 2007; Ehrsson, Spence, & Passingham, 2004; Haans et al., 2008;

Maselli & Slater, 2013; Slater et al., 2010). In this scenario of relatively high ownership, active control did not further enhance the sense of ownership (*full arm static* compared to *full arm dynamic*). However, in conditions of morphological incongruence (i.e. *detached hand static*), in which evidence of the sense of ownership was lowered, active control provided evidence in favor of the sense of ownership in the *detached hand dynamic* condition. The results reported here suggest that there is interplay between the existing level of ownership and the effect of active control. The analysis of the subgroups further strengthens this idea. Since subjects with low individual ownership scores in the *detached hand static* condition (individual mean ownership score ≤ 3) reported significantly higher scores in the *detached hand dynamic* condition.

We also assessed the interplay between visual appearance and visuomotor information on sense of agency. In the absence of visuomotor information, a sense of ownership has been shown to elicit a weak sense of agency (Kalckert & Ehrsson, 2014), as a result of congruent morphological, proprioceptive, and visuotactile information. However, when body continuity is broken, the vicarious agency significantly decreased in the absence of voluntary motor commands (Tieri et al., 2017; 2015a, 2015b). We here report the same finding in the static conditions: the reported sense of agency was significantly lower in *detached hand* compared to the *full arm static* condition in the absence of movement. However, this is not the case under active control of the movements of the virtual hand, as we found that participants did not change agency scores over the morphologically incongruent limb (i.e. *detached hand*) compared to the *full arm dynamic* condition. The presence of active control, through voluntary control of the movements of the virtual hand, plays a differential role on the effect of incongruent morphological appearance in the reported sense of agency.

A limitation of this study is that we did not present the subjects with a condition where they passively observed the moving virtual hand without performing any movement themselves. Previous studies found that visual information is sufficient to elicit the sense of agency when passively observing a moving virtual limb (Pezzetta et al., 2018; Tieri et al., 2015b). Furthermore, a break in the body continuity negatively affects reported vicarious agency when participants passively observe the moving virtual arm moving with a missing forearm (Tieri et al., 2015b). Our results indicate that, when active control is present, visuomotor information is weighted more heavily than visual information. The presence of active control could explain the difference found between our findings and what has been previously reported using passive observation of movement. The data from our *incongruent movement* condition can further support this hypothesis. In this condition, the mismatch between the performed and seen action is accompanied by a decrease in the reported sense of agency and the sense of ownership compared to the *full arm dynamic* condition. Interestingly, the difference in the individual ownership scores between the *full arm dynamic* and the *incongruent movement* conditions strongly correlates with the difference in the individual agency scores in the same conditions (Fig. 5B). The result shown in this figure suggests that congruent movement is important to maintain both a sense of agency and a sense of ownership over the virtual limb. On the other hand, manipulating morphological appearance does not present this correlation, especially in the static conditions (Fig. S2).

The primary goal of this study was to understand the effect of active control on the sense of ownership. Thus, we did not want to disclose to the participants that they would actively control the hand in the dynamic trials. Therefore, statements for the sense of agency needed to be different between the static and the dynamic conditions. As a result of this, another limitation in this study is that we could not compare the sense of agency between static and dynamic conditions. This comparison could help us further understand the interplay between morphological and visuomotor congruence in the sense of agency.

The outcome of the action has also been suggested to play a crucial role in the sense of agency (Caspar et al., 2015) and the sense of ownership (Wen et al., 2016). We labeled as outcome agency those questions that referred to the feeling of control over the consequence of the action (i.e. target lights turning off). We found no main effect for the statements assessing the outcome agency (Friedman test; $\chi^2 = 0.25$ (df = 2, n = 37), p = 0.88), and neither morphological (Wilcoxon signed-rank test; Z = -0.44, p = 0.65) nor movement incongruence (Wilcoxon signed-rank test; Z = 0.35, p = 0.72) affected outcome agency compared to the *full arm* condition (Fig. S3). The reported feeling of authorship over the consequence of the action in the environment is resistant to manipulations in the visual appearance of the limb or noise in the performance of the action. However, the relation between the role of congruent information of the outcome of the task and the morphological appearance of the limb in reported ownership needs to be further explored.

The results reported here are consistent with a Bayesian framework previously proposed to explain the mechanisms underlying the RHI paradigm (Kilteni et al., 2015; Samad et al., 2015). In this model, the sensory information is integrated to estimate whether the sensory information originates from one common source (i.e. virtual hand, which leads to the illusion of ownership) or two (i.e. virtual and real hands). We propose a similar model to qualitatively explain the results reported in this study. In our scenario, the probability that there is a single hand is estimated given the following information: morphological appearance of the virtual hand (i.e. shape of arm, body continuity, texture, color), proprioceptive information from the real hand and congruent static position of the virtual hand, and consistent information from motor feedback of the real hand and visual input of the virtual hand during movement (see Supplementary Text 1 for mathematical formulation).

In the absence of visuomotor information (i.e. *full arm static* condition), the probability that there being single hand is estimated based on the morphological appearance and proprioceptive information. With a realistic appearance of the hand and consistent proprioceptive information, we expect a high probability of estimating that there is a single hand. The results reported in the *full arm static* condition are consistent with this prediction. By introducing a break in body continuity, we expect the probability that there is a single hand to decrease. This is consistent with a reduction of reported ownership in the *detached hand static* condition. This prediction changes when the subject actively controls the movements of the virtual hand in the dynamic conditions. The additional evidence, provided by the visuomotor information, increases the probability of integrating all the information as arising from the virtual hand when compared to the static case. This is consistent with what we observe in the dynamic conditions, as the comparison between *full arm dynamic* and *detached hand dynamic* did not yield a significant decrease in the reported ownership. According to the model, the highest probability of sensory input corresponding with a single hand should take place without a break in the body continuity and under active control of the arm. Experimentally, however, in our results, we did not find an increase in the *full arm* conditions. One possible explanation is that there is a saturation effect in which ownership scores cannot be manipulated to be higher given the experimental conditions.

An analogous reasoning can be applied to compare model and experimental results for the sense of agency. The belief that the participant caused the movements of the virtual arm can also be estimated from these sensory inputs. In the absence of any motor information, we would expect congruent morphological appearance and the proprioceptive information to elicit a strong sense of agency. This is what we report in the *full arm static* condition. When a break in the continuity of the arm is introduced, we expect the probability of all the information arriving from one hand to be reduced, as we see in the *detached hand static* condition. As in the predictions for sense of ownership, this is not the case in the presence of active control. Further support for this model comes from the *incongruent movement* conditions. We would expect that the noise added to the trajectory of the virtual hand would decrease evidence in favor of a single hand, consistent with a reduction of ownership and agency scores in the *incongruent movement* condition. While at this point the model does allow for a more quantitative test, there is a qualitative agreement that suggests further tests might be fruitful.

Taken all together, our results suggest that the contribution to the sense of ownership and sense of agency from each sensory modality varies depending on the experimental context and the available information in each condition.

Author contributions

VBR and GGdP designed the project; VBR, IC, and GGdP designed the experiment; GGdP supervised project; IC and VBR programmed the experimental software; VBR collected the data; VBR and GGdP analyzed the data; VBR and GGdP wrote the manuscript.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.concog.2019.04.003.

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